

Austin Ignacious Philip

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Education

BSc, Major in Computing Science

Simon Fraser University, Burnaby, Canada

Awards: SFU Open Scholarship x 2

Expected Graduation 2028

Experience

Events Coordinator | SFU MechKeys (Student Organization)

Sep 2024 - Present

- Coordinated 11 club events (30+ avg. attendance), building custom tournament platform for registrations/scoring.
- Designed PCBs/cases for macropad workshop, teaching students keyboard building/programming fundamentals.

Frontend Developer | 8px Studio

Apr 2024 - Aug 2024

- Developed bus tracking app prototype for 50+ parents/5 admins using GPS for live updates.
- Built cross-platform iOS/Android apps with Flutter/Dart, cutting dev time 40% vs. native (4 weeks).

Data Analyst | FedEx Service Center

Apr 2023 - Jun 2023

- Automated "Shipment Data Processing" with Excel Scripts, cutting manual reporting time and boosting efficiency.
- Monitored 10,000+ delivery records via FedEx Dataworks, ensuring accurate tracking during 5-week internship.

Projects

Predictive Research Estimating Heart Disease | python, pandas, scikit-learn, shap

Apr 2026

- Built interpretable **ML pipeline in Python to predict heart disease risk**, using SHAP and feature importance to explain clinical predictions.
- Engineered data preprocessing and classification workflows with pandas and scikit-learn to evaluate risk models on clinical datasets.

Custom ESP32 Macropad | C++, Arduino-ESP32 core, TinyUSB, KiCad

Mar 2026

- Designed custom PCB in KiCad around ESP32 with USB-C and switch matrix giving a fabrication-ready board with regulated power delivery.
- Developed **C++ firmware on Arduino-ESP32** to run USB HID key emulation and real-time switch scanning enabling low-latency macro input with programmable layers.

My DormChatroom | C++, AWS

Oct 2025

- Developed a private dormitory chatroom desktop application for 25+ users, enabling real-time communication between dormmates.
- Built a TCP server hosted on AWS for automatic client connection and seamless startup, supporting 20+ concurrent users with sub-300ms message delivery.
- Designed a responsive wxWidgets UI that improved message visibility and server status monitoring, reducing user setup time by over 60%.

Meteion | Next.js, Node.js, Express.js, Socket.io, mjs

Oct 2025

- Built a **real-time drawing chat app** where users communicate through sketches instead of text.
- Integrated Google Gemini API to interpret drawings and convert visual input into text messages.
- Developed full-stack system using Next.js for frontend and Express.js with Socket.io for server management.

NeuraXtension Website Backend | Express.js, PostgreSQL, TypeScript, Drizzle

Sep 2025

- Developed **RESTful API** connecting SFU NeuraXtension frontend to CMS, enabling real-time sync and cutting manual update time.
- Designed **Supabase PostgreSQL schema** with 4+ relational tables for club events and announcements.
- Implemented type-safe Drizzle ORM layer, reducing query response times by 30% over traditional ORMs.

Skills

Languages: Python, C, C++, C#, Java, Rust, JavaScript, Dart, Godot Script

Programs/Tools: VSCode, Git, GitHub Actions, Docker, GoogleMLToolkit, AWS, Unity, Godot, Playwright, Selenium, Railway

Libraries/Frameworks: PostgreSQL, MongoDB, Node.js, MySQL, React, Next.js, ExpressJS, TypeScript, Angular, Flutter